BIDYA BHUSHAN NANDA

Full-Stack Engineer | React Native, Node.js, AI Integration

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SUMMARY

Full-stack engineer building mobile and backend systems. Develops HOSHCARE, AB Connectz's health-records app (in development), and founded Broccoli, an offline-first nutrition app newly live on Google Play (about 100 users, 5-star rated from 20+ reviews, growing quickly). Also builds local multi-agent AI tooling; electronics background (MAIT, 2029).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Rust, C/C++, SQL
Mobile & Frontend: React Native, Expo, React, Next.js, Tailwind CSS, Three.js, WebGL
Backend & Data: Node.js, NestJS, Express, Prisma, Drizzle, REST APIs, JWT auth, PostgreSQL, SQLite, Supabase (RLS)
AI & Tools: Gemini API, LLM document extraction, multi-agent orchestration, local LLMs (GGUF), Git, Docker, EAS Build, Playwright

EXPERIENCE

AB Connectz — Remote Jun 2026 – Present
Full-Stack Engineer, Paid Apprenticeship (Intern, Jun – Jul 2026) Jul 2026 – Present
HOSHCARE: health-records app I develop (in development; closed-source; walkthrough on request).

- Own the React Native / Expo app (~19 screens); review-before-save and permanent record locking ensure extracted data is never written without user confirmation.
- Built an AI pipeline that turns handwritten medical documents into structured JSON (schema-constrained output, retries, uncertainty flag for illegible text), tested on 500+ varied documents from lab reports and prescriptions to false data; fixed a schema that made the model return empty objects.
- Built the original REST API (Node.js, Express, Prisma, PostgreSQL), then moved the client to the team's shared NestJS backend and removed Firebase Auth.
- Implemented the web app on Supabase with row-level security, verified so no user can reach another's records or storage.
- Shipped the 8-page marketing site (Three.js, WebGL); fixed a silent auth failure and an Expo upload bug.

Broccoli — Founder & Developer (live on Google Play) Aug 2026 – Present

- Founded an offline-first nutrition tracker: about 100 users, 5-star rated (20+ reviews), growing quickly (details below).

PROJECTS

Broccoli — Offline-First Nutrition Tracker Aug 2026 – Present
React Native (Expo SDK 57), TypeScript, SQLite, Drizzle, NestJS, PostgreSQL | bidyabhushan.in/broccoli

- Shows totals as honest ranges, not false-precision numbers; works offline on SQLite, with South Asian food coverage.
- Built AI food recognition, barcode and label capture, 60 FPS dial interactions (Reanimated) and Skia charts with an adaptive TDEE model.
- Security-audited my own auth flow, found a deep-link password-reset token hijack risk and specified the verified App Links fix.

Habanero — Local Multi-Agent AI Engineering Engine Nov 2026 – Present
TypeScript, Python, C11, Playwright, SQLite FTS5, local LLMs (GGUF) | bidyabhushan.in/projects/habanero

- Fully offline on Apple Silicon: a planner agent splits tasks into work orders run in dependency order on local LLMs, with SHA-256 contract freezing between agents.
- 8-gate verification (Clang C11 builds, layout checks, Playwright audits) with a capped self-repair loop; passes 25/25 tasks on my own benchmark suite.

Zucchini Linux — Bare-Metal ARM64 Bring-Up (Samsung SM-T295) Oct 2026 – Present
C, Linux kernel, LK2ND, Arch Linux ARM, LineageOS | bidyabhushan.in/projects/zucchini-linux

- Dual-booted Arch Linux ARM and LineageOS on a locked-down tablet via S-Boot reverse-engineering and the LK2ND bootloader; root-caused a vbmeta bootloop. Research paper: bidyabhushan.in/blog/sm-t295-linux-bringup

Kestrel — Custom Drone Ground Control Station Oct 2026 – Present
Rust, Tokio, Tauri v2, Svelte 5, MAVLink v2, PX4, ArduPilot | bidyabhushan.in/projects/kestrel

- Cross-platform ground station with flight-critical MAVLink ingestion isolated in a Rust core; normalizes PX4 and ArduPilot flight modes and blocks arming on stale sensor data.

EDUCATION & ACTIVITIES

B.Tech, Electronics Engineering — MAIT, Delhi 2025 – 2029 (expected)
CGPA 8.5 / 10 | Aerostars UAV Society member | Paper "Augmentation, Not Obsolescence" at bidyabhushan.in